

APRIL 2024

2408325 / UMSA21I

Time: Three Hours

Maximum Marks: 75

SECTION A – (10 × 2 = 20 marks)

Answer the following questions

1. Find the value of $I_n = \int_0^{\frac{\pi}{2}} \cos^n x \, dx$
2. Integrate $\int \cos^4 x \, dx$
3. Define Even function and give some examples.
4. Define Fourier series
5. Solve $p^5 + py = 1$
6. Find the general solution of the differential equation $y'' + 2y' - 8y = 0$
7. Find the inverse Laplace transform of $\frac{s^2}{(s^2+a^2)(s^2+b^2)}$
8. Find the Laplace transforms of $\sin t \sin 2t \sin 3t$
9. Evaluate $P^2 + Pxy = y^2 10gy$.
10. Find ∇r .
11. What is Solenoidal vector?
12. Find a unit normal to the surface $x^2y + 2xz^2 = 8$ at the point (1,0,2).

SECTION B – (5 × 5 = 25 marks)

Answer any FIVE questions

13. Integrate $\sin^8 x \cos^5 x \, dx$ with respect to x .
14. Find the fourier series of $f(x) = x$ in $(0, 2\pi)$ of peroidicity 2π .
15. Find the value of $\int_0^{2a} \frac{x^3 dx}{\sqrt{2ax-x^2}}$.
16. Solve $p^2 + q^2 = npq$.
17. Find the Laplace transform $\frac{1-e^t}{t}$.
18. Find the inverse laplace transform of $\left\{ \log \left(1 + \frac{\omega^2}{s^2} \right) \right\}$.
19. Prove that $\text{curl} (u \times v) = (v \cdot \nabla)u - (u \cdot \nabla)v + u \, \text{div} \, v - v \, \text{div} \, u$.

SECTION C – (3 × 10 = 30 marks)

Answer any THREE questions

20. Evaluate i) $\int \tan^3 x \, dx$ ii) $\int \tan^4 x \, dx$.
21. Find the fourier series of $f(x) = x^2$ in $(-\pi, \pi)$ of peroidicity 2π .
22. Solve $z^2(p^2x^2 + q^2) = 1$.
23. Evaluate $L^{-1} \left[\frac{3s^2+16s+26}{s(s^2+4s+13)} \right]$.
24. Prove that $\nabla \cdot (F \times G) = G \cdot (\nabla \times F) - F \cdot (\nabla \times G)$.